IRB Administration Office
87 E. Canfield, Second Floor
Detroit, MI 48201
www.research.wayne.edu/irb

Notice of IRB Administrative Determination: Non-Human Participant Research (HPR)

To: Gregory Schwing
From: Helen Craigie D.Bioethics
Date: April 21, 2026
Re: Harnessing Algorithmic Disagreement for Automated Lumbosacral Transitional Vertebrae Detection

WSU IRB HPR Number: **2026 144**

Materials in the above referenced proposal were received by the Wayne State University Institutional Review Board Administration Office via Human Participant Research (HPR) Determination Tool on 04/21/26

This project does not constitute human participant research according to the definition codified in the Common Rule at 45 CFR 46 and FDA regulations. This means that IRB review and oversight is not required for this project. We recommend you keep this memo for your records.

The project also does not involve individuals who would receive a test article (drug or device) as participants and therefore the FDA regulations do not apply. Thus, this project does not require review or approval by the Wayne State University Institutional Review Board.

Please be aware that projects involving identifiable Personal Health Information (PHI) which the IRB has determined does not meet the definition of human participant research must still adhere to HIPAA regulations, and may require authorization from the covered entity's privacy officer*.

Projects must also obtain support from any unaffiliated institution(s) involved.

Please contact applicable affiliates: (DMC, Karmanos Cancer Institute, J.D. Dingell Veterans Administration Veterans Medical Center) to obtain authorization to conduct your project.

Please note that changes to the study plan may impact this determination as to whether the project constitutes Human Participant Research. Please contact the IRB Administration office if there are changes to the study plan that may affect this determination.

Helen Craigie Digitally signed by Helen Craigie
Date: 2026.04.21 10:05:15 -04'00'

Wayne State University IRB Administration Office

Human Participant Research Determination Tool

According to federal regulations from the Department of Health and Human Services (DHHS) and the Food and Drug Administration (FDA), all human subject/participant research must be reviewed and approved by the Institutional Review Board (IRB). This determination tool is designed to determine whether an activity is considered human participant research under either regulation. The IRB Administration Office can provide assistance with making this determination.

Human Participant Research (HPR) Guidance:

Before completing this determination tool, we recommend reviewing the guidance available to help understand how the IRB applies the regulatory definitions to the HPR Determination reviews:

- Does My Study Need IRB Review: Part 1- On- Demand video
- Does My Study Need IRB Review: Part 2- On- Demand video
- HPR Guidance Document

If you determine that the project does not require IRB review:

If by the use of this tool, you have determined that the project does not require IRB review, you do not need to submit this form to the IRB office unless you need documentation from the IRB to concur with your self-determination. Add the project title and name of the person completing the project, their title and the date the tool was completed to the first page, and retain this tool in your files to document this determination.

If there are any modifications to your project that could change this determination, please complete this tool again. Submit the appropriate application to the IRB if changes to your project result in the determination that human participant research is involved according to this tool, which is based on the federal regulations.

If you determine that IRB review is required:

If you have determined that IRB review is required, IRB approval must be obtained **before** conducting any human participant research activities. Visit the WSU IRB website for additional information and the forms required for a new submission: www.research.wayne.edu/irb . E-mail irbquestions@wayne.edu with any questions that come up along the way.

If assistance is needed, or if written documentation from the IRB office is required:

Complete the **entire** form and email it to IRBQuestions@wayne.edu. Include any relevant supporting documents (e.g., grant, protocol, data collection tools). Please do not submit handwritten documents to the IRB office.

IRB Administration Office
87 E. Canfield, Second Floor
Detroit, MI 48201
Telephone# (313) 577-1628
www.research.wayne.edu/irb

IRB Determination Number (IRB Use Only)

2026 144

Section A: Project Staff and Location Information:

Complete all sections of this form if you will be requesting assistance from the IRB Administration Office in making your Human Participation Research determination. Otherwise continue on to Section D to begin the determination tool.

Project Title:	Harnessing Algorithmic Disagreement for Automated Lumbosacral Transitional Vertebrae Detection			
Name of person conducting the project:	Gregory Schwing	Title:	PGY-1 Resident; MD-PhD Candidate	Date: 04/21/26
Email Address:	go2432@wayne.edu	Phone:	[redacted]ld]	
Status: **Select all that apply**	☐ Wayne State Faculty ☑ WSU Graduate Student ☐ WSU Undergraduate Student ☑ Resident/Fellow/Trainee ☐ Other: ☐ DMC Staff ☐ Karmanos Staff ☐ J. D. Dingell VAMC Staff			
Division or College:	School of Medicine	Department:	Orthopedic Surgery	
Campus Address:	540 E. Canfield St, Detroit, MI 48201			
Faculty Sponsor/ Supervisor for this Project:	Name: Rahul Vaidya ☐ I do not have a Faculty Sponsor/Supervisor	Title: Phone:	Chair, Orthopedic Surgery [redacted]ld]	
Form completed by:	Gregory Schwing	E-mail:	go2432@wayne.edu	
Check ALL that apply:	☐ Behavioral, social, education, non-medical study ☑ Medical study			

Useful Tips:

1. Carefully read through the definitions provided within the determination tool prior to answering the questions. This will help to ensure an accurate determination is made.
2. If your research involves the use of de-identified data or bio-specimens, include a letter of support from the institution/department providing the data/bio-specimens that confirms that the data/bio-specimens provided will be stripped of all identifiable information before you receive it.
3. If your project requires you to **access**, or **use** identifiable data to complete your data collection, you should check yes to #2.2 in Section D even if the data you are retaining/collecting for analysis will be de-identified.
    a. For example, accessing a patient's medical record to collect data for a chart review gives you access to individually identifiable data even though you have no plans to retain/collect any identifiable data. This is considered a human participant as described by #2.2 under section D of this tool.
4. Review the guidance available on the cover page

Section B: Location and Study Data Details:

1.	Describe the location(s) where activities will take place: All activities are computational and are performed on the Wayne State University Warrior High Performance Computing cluster. No human-subject contact, recruitment, or in-person data collection occurs at any location.		
2.	Select the type of data being collected: Check all that apply:	☑ Secondary or retrospective data collection *(Existing data that was originally generated for reasons unrelated to this project)* ☐ Identifiable Protected Health Information (PHI) ☐ Prospective Data ☐ Secondary or retrospective collection of bio-specimens ☐ Prospective collection of bio-specimens	
3.	Will you require **access** to, or the **use** of any identifiable information pertaining to participants during the process of collecting data?	☐ Yes ☑ No	
4.	Will data being obtained include individually identifiable information?	☐ Yes ☑ No	If yes, list all identifiers being collected (e.g. name, date of birth, medical record number, email address, other codes; etc.): None. The dataset is fully de-identified by the data provider (Radiological Society of North America) prior to public release; no names, dates of birth, medical record numbers, dates of service, geographic identifiers, or other HIPAA identifiers are obtained at any stage.

5.	Could the identities of participants be known to, or be readily ascertained by the investigators?	☐ Yes ☑ No
6.	Select the source of data collection: Check all that apply	☐ Medical Record Review: Complete #6a. ☐ Survey ☐ Interview ☐ Bio-bank ☐ Data Repository ☑ Other Describe: Public deep-learning challenge dataset: RSNA 2024 Lumbar Spine Degenerative Classification, distributed by the Radiological Society of North America through Kaggle (https://www.kaggle.com/competitions/rsna-2024-lumbar-spine-degenerative-classification).
6a.	Indicate the institution(s) (Covered Entity) you will be reviewing and collecting medical record data from: Check all that apply:	☐ Detroit Medical Center Facility ☐ Karmanos Cancer Institute ☐ J.D. Dingell Veterans Administration Medical Center ☐ Other Describe: ☑ N/A- Project does not involve the review of or collection of data from a medical record
	Note: All applicable institutional policies must be followed at all times. When collecting medical record data outside of your normal responsibilities within the institution, you must obtain approval to access and collect medical record data.	

7.	Describe how the data will be obtained (e.g. survey, interview, observation, testing, review of existing records, etc.): Data are obtained by direct public download of the RSNA 2024 Lumbar Spine Degenerative Classification dataset from Kaggle (https://www.kaggle.com/competitions/rsna-2024-lumbar-spine-degenerative-classification) under the competition's public terms of use, which permit academic and research use. The dataset was de-identified by the Radiological Society of North America prior to public release. The investigator accessed the dataset under the same public terms available to any researcher and did not receive any additional identifiers, linking keys, or supplemental PHI from any source.

Section C: Description of the Project:

Provide a description of the project with enough detail for the determination.

8.	Describe the purpose of this project: To evaluate whether labeling disagreement among independently developed deep-learning spine-segmentation models can serve as an automated detection signal for lumbosacral transitional vertebrae (LSTV) on lumbar MRI. LSTV is a common congenital variant (estimated prevalence 4-35%) that is frequently overlooked on routine imaging and is a recognized cause of chronic low back pain and surgical mis-leveling. Manual identification is observer-dependent and inconsistent across radiologists; an automated, model-agnostic screening signal could improve detection in both research and clinical workflows without requiring additional patient data.

9.	Describe the objectives or aims for the project: (1) Apply three independently developed, publicly available deep-learning spine-segmentation models — TotalSpineSeg, SPINEPS/VERIDAH, and the Ian Pan disc localizer — to publicly available lumbar MRI exams from the RSNA 2024 Lumbar Spine Degenerative Classification dataset. (2) Quantify per-vertebra and per-disc labeling disagreement among the three models. (3) Test whether disagreement patterns and vertebral-angle features (per Seilanian Toosi 2025) correlate with the presence and Castellvi subtype of LSTV. (4) Develop a Bayesian probability framework integrating the disagreement signal with vertebral-angle features to produce a calibrated automated LSTV detection score.
10.	Describe how the results will be used/applied: Findings will be presented at the 43rd Annual Meeting of the American Association of Clinical Anatomists (Mayo Clinic, Rochester MN, June 2026) and prepared for submission to a peer-reviewed clinical-anatomy or imaging-AI journal. The validated detection signal is intended to support downstream automated screening tools that operate on standard clinical lumbar MRI without requiring patient-specific identifiers, institutional data access, or additional patient interaction.

11.	Describe the participants for the project: No human participants are recruited, contacted, interviewed, or directly observed at any stage of this project. The analysis uses only fully de-identified imaging exams from the publicly released RSNA 2024 Lumbar Spine Degenerative Classification dataset, distributed through Kaggle under the competition's public terms of use. The dataset was de-identified by the Radiological Society of North America prior to public release. The investigator has not received and cannot obtain any identifiers, linking keys, dates of service, or other PHI associated with the imaging studies. The investigator is not affiliated with any of the institutions that originally generated the imaging studies and has no means of re-identifying any individual whose data appear in the dataset.

Section D: Determination Tool:

In order for a project to meet the regulatory definition of human participant (subject) research under HHS regulation 45 CFR 46.102 and FDA regulation 21 CFR 50.3, the project must meet the definition of a human participant(subject) and the definition of research. The following questions will help to determine if your project meets these definitions and requires IRB review.

Step 1: Does the Project Meet the Regulatory Definition of Research Under HHS Regulations (45 CFR 46)?

Research means a systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge.

Select all that apply:

1	1.1	☑	Information is expected to expand the knowledge of a scientific discipline or other scholarly field or study and yield one or both of the following:	
			☑	Results are intended to be applicable to a larger population beyond the site of data collection or the specific participants/subjects studied.
			☑	Results are intended to be used to develop, test, or support theories, principles, and statements of relationships or to inform policy beyond the study.
	1.2	☑	The information is collected to share with others in a discipline and is created to make a broad statement (conclusion) about a group of people, procedures, programs, etc.	

Step 2: Does the Project Involve Human Participants/Subjects Under HHS Regulations (45 CFR 46)?

Human subject means a living individual about whom an investigator (whether professional or student) conducting research:

- Obtains information or bio-specimens through intervention or interaction with the individual, and uses, studies, or analyzes the information or bio-specimens; or
- Obtains, uses, studies, analyzes, or generates identifiable private information or identifiable bio-specimens

2.	2.1	Does the research involve collecting data through *intervention or **interaction with the individuals?	☐ Yes: ☑ No:
		Definitions (45 CFR 46.102): ***Intervention includes:*** • Both physical procedures by which information or bio-specimens are gathered and manipulations of the participant or the participant's environment that are performed for research purposes. ***Interactions involve:*** • Communication or interpersonal contact between the investigator and the participant. Interactions can be in-person or virtual. Examples include interactions via email or social media, or by completing a survey, questionnaire, interview or participation in a focus group.	
	2.2	Will you obtain, use, study, analyze, or generate *identifiable private information or **identifiable bio-specimens from individuals?	☐ Yes: ☑ No:
		Definitions (45 CFR 46.102): ***Identifiable private information:*** information for which the identity of the participant may be readily ascertained by the investigator or associated with the information. **Private information includes:**	

			• Information which has been provided for specific purposes by an individual and which the individual can reasonably expect will not be made public (e.g., a medical record, emails, certain listserv communications, class papers and exams, etc.) • Identifiable private information is private information for which the identity of the subject is or may readily be ascertained by the investigator or associated with the information. ****Identifiable bio-specimen** is a bio-specimen for which the identity of the subject is or may be readily ascertained by the investigator or associated with the bio-specimen.

Step 1 & Step 2 Human Participant Research Determination Decision Tool:

Step 1: Research Determination:

- If you checked any boxes in question 1.1 or 1.2, the project is considered research per HHS (45 CFR 46.102(l))

Step 2: Human Participant Determination:

- If you answered yes to question 2.1 or 2.2, the project involves human participants per HHS (45 CFR 46.102 (E))

Human Participant Research Determination:

The project is HPR and IRB Review is required if:	Step 1 finds that your project involves research **and** step 2 finds that your project involves human participants**.** • **Submission instructions and guidance are available on the IRB website:** www.research.wayne.edu/irb. • Email your questions to irbquestions@wayne.edu
The project is not HPR and does not require IRB review if either of the following applies:	• If step 1 finds that your project does not involve research and/or • If step 2 finds that your project does not involve human participants
Step 3: Proceed to Step 3 to determine if the project is subject to Federal Drug Administration (FDA) regulations.	

Step 3: Does the activity require IRB review under the FDA Regulations (21 CFR 50)?

Select all that apply

3.	3.1	☐	Activity is conducted in the United States and involves the **use of a drug** in one or more *human subjects (as recipients of a **test article or as controls, patient or healthy 21 CFR 50.3) but is **not** the use of an approved drug in the course of medical practice
	3.2	☐	Activity is conducted in the United States and evaluates the safety or effectiveness of a **device** in one or more *human subjects.
	3.3	☐	Data regarding *human subjects (including controls) will be submitted to or held for inspection by FDA as part of an application for a research or marketing permit.

	3.4	☐	Data regarding the use of a device on human specimens (including de-identified/anonymous) specimens will be submitted or held for inspection by the FDA as part of an application for a research or marketing permit

Guidance- Definition (21 CFR 50.3)

***Human Subject:** an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient.

****Test article:** any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other article subject to regulation under the Food, Drug, and Cosmetic Act.

FDA Human Subject Research Determination Instructions:

If any of the boxes in #3 are checked, the activity is human research per FDA regulations and subject to IRB review.

Note: If the activity is determined to be human research subject to FDA regulation, this does not necessarily mean it is also subject to HHS regulation, and vice versa. Separate determinations should be made.

If assistance is needed, or if written documentation from the IRB office is required:

Complete the **entire** form and email it to IRBQuestions@wayne.edu. Include any relevant supporting documents (e.g., grant, protocol, data collection tools). Please do not submit handwritten documents to the IRB office.

WSU IRB Determination:

(To be completed by IRB Administration)

☑	**Not Human Participant Research - IRB review is not required**	
	Type of activity that does not require IRB review:	
	☐	Case Report **Note:** IRB approval is required if the case report involves more than three cases
	☐	Course Related Activities **Note:** IRB Approval is required if a student is involved in an activity designed to teach research methodologies and the instructor or student wishes to conduct further investigation and analyses in order to contribute to scholarly knowledge.
	☐	Decedents: Research limited to death records, autopsy materials or cadaver specimens. **Note:** IRB approval is required if decedent information contains psychotherapy notes, or information related to HIV, mental health, genetic testing or drug or alcohol abuse
	☐	Journalism/Documentary Activities **Note:** IRB approval may be required when journalists conduct activities normally considered scientific research intended to develop generalizable knowledge (e.g. systematic research, surveys, and/or interviews that are intended to test theories or develop models).
	☐	Oral History **Note:** IRB approval is required when the activities are intended to develop generalizable conclusions (e.g., that serve as data collection intended to test economic, sociological, or anthropological models/theories)
	☐	Program Evaluation/Quality Improvement/Quality Assurance Activities **Note:** Investigators conducting QI/QA projects should ensure that they have received approval from any applicable committees within their department or the site in which the activity will occur
	☐	Public Use Datasets **Note:** IRB approval is required for the use of restricted use data, if a proposal is required to obtain the dataset, or if a data use agreement is involved.
	☑	De-Identified Private Information and/or Human Biological Specimens **Note:** IRB approval is required if the information being collected could enable the investigator to identify or readily ascertain the identity of the individual whom the private information or specimens belongs to.
	☐	Public Health Surveillance **Note:** Activities are limited to those necessary to allow a public health authority to identify, monitor, assess, or investigate conditions of public health importance
	☐	Other Project type not considered to be HPR Describe:

☐	**Project is Human Participant Research- IRB Review Required.**	
	Type of IRB required:	
	☐	Exempt IRB review is required (minimal risk research): Rationale:
	☐	Expedited IRB review is required (minimal risk research): Rationale:
	☐	Full Board IRB review is required (greater than minimal risk research): Rationale:

Reviewer Comments:	Research: The project meets the regulatory definition of research as it is likely to contribute to generalizable knowledge (45 CFR 46.102(l)). Human Subjects/participants: This project does not meet the regulatory definition of human subjects/participants (46 CFR 46.102(e)) because investigators will be analyzing de-identified data in a manner that will make it impossible to ascertain the identity of the individual's data being analyzed for research.
Authorized IRB Reviewer Signature:	Helen Craigie Digitally signed by Helen Craigie Date: 2026.04.21 10:03:52 -04'00'

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